When Structure Is Not Utility: Localizing Failure Modes of Structural Interventions in Unsupervised Clustering

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Abstract

Structural signals are widely used to shape unsupervised representations, but a useful relation is not necessarily a useful intervention. We study this gap as a mechanism-localization problem rather than as a search for another architecture. The audited V1–V22 history contains 2,209 registry rows, 1,637 paired Delta ARI records, and 431 dataset/protocol/readout units; its 194 material-positive, 680 material-negative, and 763 observed-small rows form an observational Failure Atlas. We then analyze a predeclared V21 case study with a shared warmup branchpoint and budget-, donor-, eligible-set-, batch-, and selection-noise-matched None, Random, and Topology policies. The primary per-seed endpoint is $S_{\mathrm{full\_ARI}} = \mathrm{ARI}_T - \mathrm{ARI}_R$; the reported S_d values are the dataset means of those three seed-level endpoints. The confirmation means are $+0.044617$ for Baron Human and -0.065332 and -0.033410 for Campbell and `hate_speech`, respectively. Thus topology-dependent selection has a conditional, sign-heterogeneous effect in this audited case study, not universal superiority. Feature, gradient, and local/global records remain diagnostics, not universal causal explanations. A predeclared holdout produced zero of six evaluable panels; independent validation is not established, and the status is recorded as `inconclusive_not_completed`, not as a negative model result.

1 Introduction

Local neighborhoods, graphs, and related structural signals are established ways to shape unsupervised representations [1, 3]. They can provide useful inductive bias for sparse, high-dimensional observations, but a structural signal is not itself a clustering objective. The operation that selects a relation, applies a perturbation, updates a representation, and then produces a partition can preserve, distort, or discard the information available at the start.

We study the gap between *structural opportunity* and *intervention utility.* A graph may contain a locally informative relation while a learned selector chooses the wrong coordinates. A fixed-budget intervention may damage cluster-defining features. An auxiliary objective may compete with representation learning [4]. Or a local geometric improvement may fail to survive the global readout [1]. These stages are often collapsed into one score difference between a structural variant and a baseline. Such a difference shows that a variant changed a score; it does not identify whether the change came from the structural policy, the generic intervention, the optimizer, or the readout.

The problem becomes sharper when evidence accumulates across exploratory versions. Different versions may change preprocessing, intervention location, training target, effective budget, stochastic schedules, or final readout. Seeds and variants are then repeated measurements rather than independent data sets. A credible synthesis must preserve protocol and dataset boundaries while still exposing recurring patterns. It must also distinguish post-treatment descriptors from pre-intervention properties and must not turn incomplete computation into a negative result.

Our question is therefore: **why does apparently useful structural information fail to translate reliably into unsupervised clustering utility?** We use the localization chain

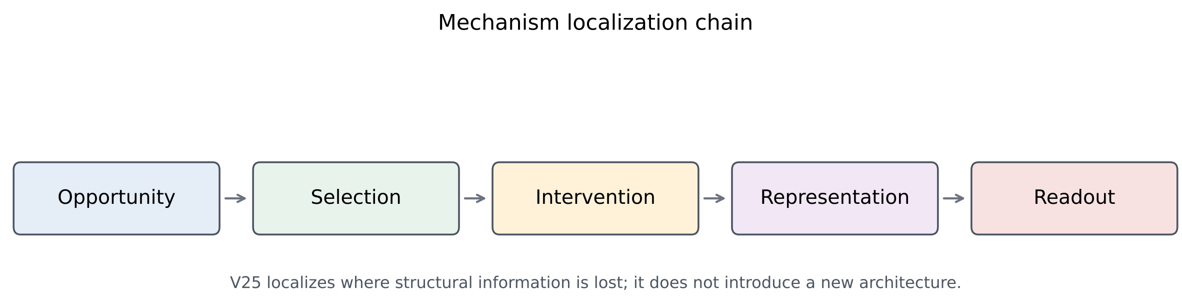


Figure 1: Frozen V25 mechanism-localization chain. The diagram is a protocol schema for separating opportunity, policy, intervention, representation, and readout; it is not a fitted architecture or a causal graph estimated from the historical rows.

$$\textbf{Opportunity} \rightarrow \textbf{Selection} \rightarrow \textbf{Intervention} \rightarrow \textbf{Representation} \rightarrow \textbf{Readout}.$$

The chain asks whether an opportunity is measurable, whether topology changes what is selected, whether a matched intervention is helpful, whether the objective moves the representation compatibly, and whether local changes survive the global readout. This is a study protocol and claim firewall, not a new neural architecture.

Our contributions are twofold. First, we provide an auditable observational Failure Atlas for the V1–V22 history, with repeated-measurement units, protocol/readout boundaries, and a separate V23/V24 boundary branch. Second, we provide a matched V21 selection-policy decomposition that separates generic intervention from topology-dependent selection and exposes conditional signs. Feature, gradient, and local/global analyses localize possible failure stages; they are not promoted to universal causal laws.

2 Study Design and Methods

2.1 Evidence layers and statistical units

The A0 registry records dataset, protocol, variant, seed, source and preprocessing hashes, readout and K provenance, structural source, selection policy, intervention location, training target, measurement timing, causal status, artifact availability, label/K isolation, reuse provenance, and alternative explanations. V1–V22 enter the quantitative atlas. V23 and V24 are registered as boundary evidence but are not pooled with structural-intervention rows.

The primary retrospective unit is dataset/protocol/readout. Seed and variant rows are repeated measurements. We do not treat the 1,637 paired records as 1,637 independent data sets. A1 uses raw Delta ARI as its primary descriptive quantity; headroom-aware summaries and intervention-magnitude descriptors are sensitivity or post-treatment diagnostics, not causal covariates.

The A1 numbers are imported historical metrics: A1 does not reload labels or recompute historical ARI values. Consequently, the atlas preserves the original evaluation and selection provenance, including the historical V21 dataset-selection bias, rather than presenting a newly harmonized label-free benchmark.

Table 1: Evidence layers and their inferential boundary.

Layer	Evidence	Statistical unit	Causal status
A0/A1	V1–V22 registry and atlas	dataset / protocol / readout	observational
A0 boundary	V23/V24 No-Go records	boundary record	isolated historical boundary
E1	V21 matched N/R/T	dataset; seeds repeated	matched prospective case study
E2	feature and optimizer diagnostics	dataset × seed	diagnostic/post-hoc
E3	local/global replay boundary	artifact-complete case	observational boundary
Phase D	predeclared holdout	six panels	incomplete compute

2.2 Failure Atlas and mechanism triage

A1 admits only artifact-complete cases to offline replay. Metadata-only snapshots remain provenance records and cannot be reconstructed into replay evidence. The atlas separates structural opportunity, baseline/headroom, local/global conversion, and post-treatment magnitude descriptors. Because no common fixed-graph/null opportunity endpoint exists across all historical protocols, the atlas is observational rather than a pooled causal test of structural quality.

A2 constructs a Claim–Evidence Matrix with historical support, counterexamples, identifiability, required generality, fatal falsifiers, alternative explanations, novelty, and required new compute. It can retain E1, cancel E1, or authorize no prospective computation; it cannot invent an E4 after triage. In the frozen artifacts A2 retained E1 and froze the selection claim, the primary per-seed endpoint $S_{\text{full\_ARI}} = \text{ARI}_T - \text{ARI}_R$, the claim-dependent holdout subset, and the practical threshold $\delta = 0.03$.

2.3 Matched V21 case study

The prospective case study compares three arms:

$$N = \text{matched None}, \tag{1}$$
$$R = \text{matched Random assignment policy}, \tag{2}$$
$$T = \text{topology-dependent selection policy}. \tag{3}$$

The estimands are

$$I_d = Q(R) - Q(N), \tag{4}$$
$$S_d = Q(T) - Q(R), \tag{5}$$
$$Q(T) - Q(N) = I_d + S_d. \tag{6}$$

For E1, Q is the clean-embedding, known-K KMeans ARI for one seed. Thus $S_{\text{full\_ARI}}$ is the primary seed-level endpoint and $S_d = \text{mean}_{s \in \{42,123,7\}} S_{\text{full\_ARI},d,s}$ is only the dataset-level summary used for the four-state rule and figures. Pilot and confirmation panels are audited and reported separately; they are not pooled.

The losses are frozen as

$$L_N = L_{\mathrm{scMAE}} + \lambda_i L_{\mathrm{InfoMax}}, \tag{7}$$
$$L_R = L_{\mathrm{scMAE}} + \lambda_i L_{\mathrm{InfoMax}} + \lambda_a L_{\mathrm{JS}}^{R}, \tag{8}$$
$$L_T = L_{\mathrm{scMAE}} + \lambda_i L_{\mathrm{InfoMax}} + \lambda_a L_{\mathrm{JS}}^{T}. \tag{9}$$

All arms start from one branchpoint after warmup and Student-t head initialization and before the first assignment update. Base reconstruction masks and donors, assignment donors, eligible coordinates, exact effective budgets, batch order, topology statistics, Gumbel selection noise, and readout randomness are shared or hash-audited. The Random arm uses zero topology logits plus the same frozen Gumbel tensor; the Topology arm adds topology-dependent logits. None preserves the reconstruction and InfoMax terms and optimizer state but executes no assignment forward and no JS loss.

The primary readout is clean embedding plus known-K KMeans. The Student-t head is a secondary diagnostic. The full label vector is excluded from preprocessing, graph construction, Gate, loss, optimizer updates, and all model state transitions. Labels enter only after fitting for outer ARI/NMI evaluation; benchmark-known K may nevertheless be derived from the outer labels to size the trainable head and primary readout. E1 is therefore a real-ground-truth, known-K benchmark, not fully label-free fitting.

The pilot contains cnae9, Mouse Retina, and SMS with seeds $[42, 123, 7]$; the confirmation panel contains Baron Human, Campbell, and `hate_speech` with the same seeds. A dataset-level state is Positive or Negative only when the mean effect exceeds ± 0.03 and at least two of three seed effects share its sign. Observed-Small requires the mean and all three seed effects to be smaller in absolute value; otherwise the state is Inconclusive. No three-seed bootstrap is used to claim equivalence.

2.4 Localization diagnostics

E2-A compares topology-selected and eligible-but-not-selected coordinates, but aggregates each metric at dataset-by-seed level. Coordinate-level distributions are descriptive plots only. Fisher separation, mutual information, and class-support enrichment are post-hoc label-aware descriptors and are not fit inputs. E2-B records base, assignment, and InfoMax gradient geometry at T0, T1, and T2. E2-C clones model, head, and Adam state and executes actual N/R/T Adam one-step updates. These diagnostics do not establish a universal objective-compatibility law.

E3 applies the artifact-complete replay gate and keeps label-free geometry separate from post-hoc supervised neighborhood metrics. The frozen replay summary has zero eligible rows (`candidate_rows=0`), so no E3 replay was run; the V23 local/global rows remain separate boundary evidence. A local improvement accompanied by non-positive ARI is therefore only a possible conversion-failure pattern, not a universal local-to-global theorem.

2.5 Holdout and closure

Before E1 outcomes, the holdout candidate pool, input adapters, normalization, feature selection, graph/model inputs, source hashes, measurement schema, and claim-dependent activation rule were frozen. The manifest records a scRNA shortfall and does not silently fill it using outcomes. Phase C selected the selection claim, so Phase D activated the N/R/T subset and primary $S_{\mathrm{full\_ARI}}$ endpoint. The expected holdout contained six panels. No panel produced an evaluable primary endpoint under the frozen dense decoder/ Adam resource path; this is incomplete compute, not an independent replication result.

Table 2: Counts in the frozen retrospective atlas. Rows are repeated records, not independent data sets.

Quantity	Value
V1–V22 registry rows	2209
Paired ΔARI rows	1637
Dataset/protocol/readout units	431
Material positive rows	194
Material negative rows	680
Observed-small rows	763

Table 3: Confirmation means for the matched V21 decomposition. Each row is a dataset mean over three repeated seeds.

Dataset	I_d	S_d	S_d state
Baron Human	-0.011463	+0.044617	Positive
Campbell	+0.132841	-0.065332	Negative
hate_speech	-0.002671	-0.033410	Negative

3 Results

3.1 Historical Failure Atlas

The registry contains 2,209 V1–V22 rows, including 1,637 paired Delta ARI rows grouped into 431 dataset/protocol/readout units. The paired atlas contains 194 material-positive, 680 material-negative, and 763 observed-small rows at $\delta = 0.03$. The signs and magnitudes vary across intervention families and protocols, but the registry does not provide one common structural-opportunity endpoint. We therefore report these values as an observational atlas rather than as a pooled causal estimate.

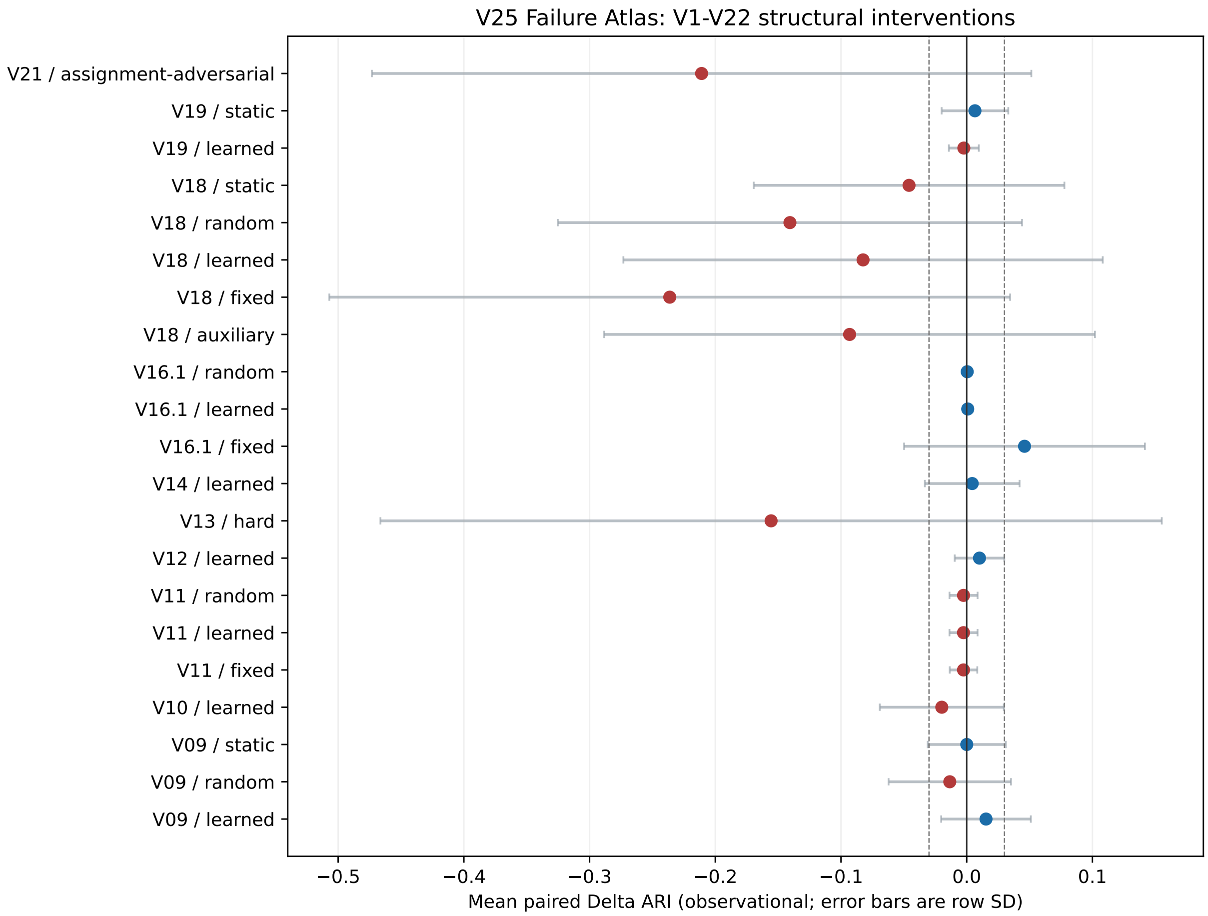
3.2 Matched selection effects

The audited E1 panels pass the full matching and label-isolation contract. The per-seed primary endpoint is $S_{\text{full\_ARI}}$; the confirmation dataset means $S_d = \text{mean}_s(S_{\text{full\_ARI}})$ are shown in Table 3 and Figure 3. The primary selection effect changes sign across the three confirmation data sets. The generic intervention effect is also heterogeneous: it is positive for Campbell, observed-small for `hate_speech`, and inconclusive for Baron Human. This decomposition separates generic intervention from the topology policy increment; it does not prove either component universal.

The pilot gate passed because at least two of three pilot datasets showed a seed-stable material effect in either I_d or S_d; signs were not required to agree. Pilot and confirmation are not pooled: the confirmation panel alone supplies the three case-study values above. The result is therefore evidence for conditional utility and heterogeneity, not a reason to search for a universally beneficial selector.

3.3 Localization diagnostics

E2 produces dataset-by-seed semantic summaries and gradient/one-step records under the frozen N/R/T protocol. These artifacts permit inspection of selected feature semantics and optimizer-direction changes, but they do not produce a stable cross-dataset objective-conflict rule. Figure 4 shows the diagnostic geometry without fitting a cross-dataset causal model.



Rows are repeated records within dataset/protocol/readout units; no pooled causal inference.

Figure 2: V1–V22 Failure Atlas. The display is observational and stratified by version/family; it is not a pooled causal estimate.

The E3 replay gate found zero artifact-complete candidate rows, so no replay was run. The separate V23 boundary rows are retained only to show why local and global readouts must be measured separately; they do not support a universal conversion law.

3.4 Independent validation status

The holdout has no evaluable endpoint: zero of six panels completed under the frozen protocol, and the recorded cause is the dense decoder/Adam resource boundary. Independent validation is therefore not established. We do not use this incomplete compute to falsify the selection claim or report a negative performance result.

4 Discussion

The central result is a separation between structural opportunity and structural intervention effect. In the broad history, heterogeneous signs appear across protocols and readouts, but the historical data cannot identify one common opportunity treatment. In the narrower V21 counterfactual,

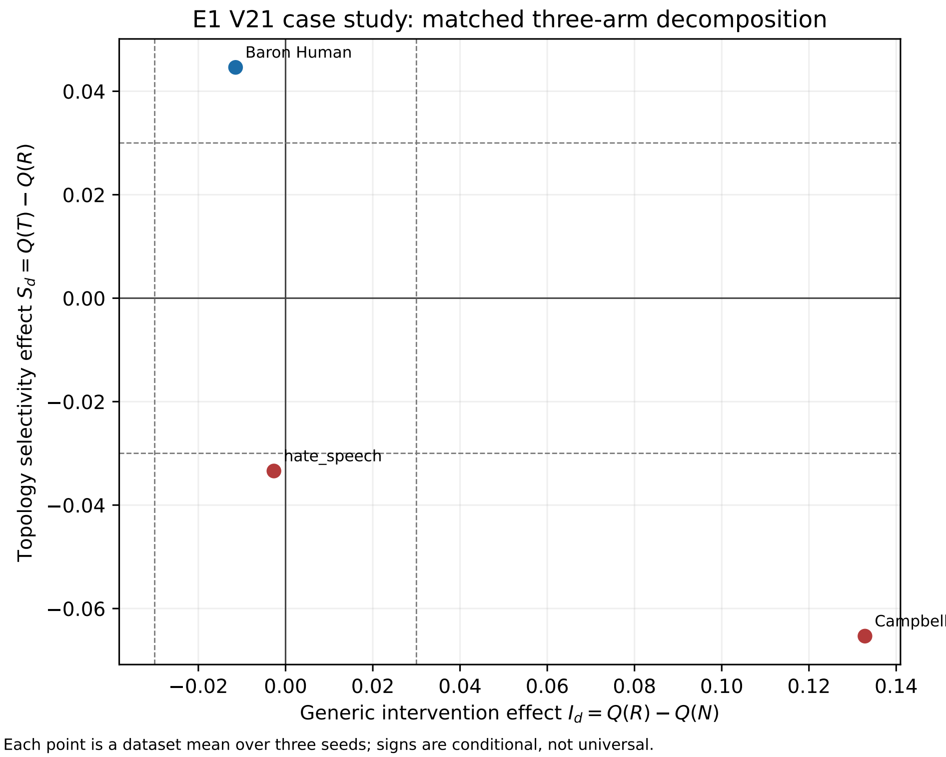


Figure 3: Generic intervention effect I_d versus topology selectivity effect S_d. The signs are conditional case-study evidence, not a population claim.

topology-dependent selection still changes sign after generic intervention and selection noise are matched. The defensible conclusion is a conditional, sign-heterogeneous selection effect: how and where structure intervenes matters at least as much as whether a structural signal can be measured.

This result changes what a next method would need to demonstrate. A more elaborate selector is not justified merely because a previous selector failed. A convincing method would need to show, under a matched generic-intervention control, that its policy changes the primary readout in a stable direction. It would also need to separate label-free training evidence from post-hoc semantic diagnostics. The current study does not propose that method; it supplies the failure-localization protocol needed before designing one.

The findings also explain why local geometry, reconstruction difficulty, and assignment stability should not be used interchangeably. A local metric can be post-treatment and label-aware [1], a reconstruction loss can reward a difficult coordinate without improving the partition [4], and a training head can disagree with a clean-embedding KMeans readout. Treating these as the same utility signal would recreate the experimental loop this study is designed to stop [2].

5 Limitations and Conclusion

First, the historical atlas has no common structural-opportunity endpoint. Its value is breadth and failure coverage, not causal identification across all versions; A1 imports historical metrics and retains the historical V21 dataset-selection bias. Second, the prospective evidence is a six-dataset V21 case study with known-K evaluation; its conditional signs cannot be generalized to every structural intervention or domain. Third, E2 and E3 are diagnostics, with dataset-by-seed aggregation and

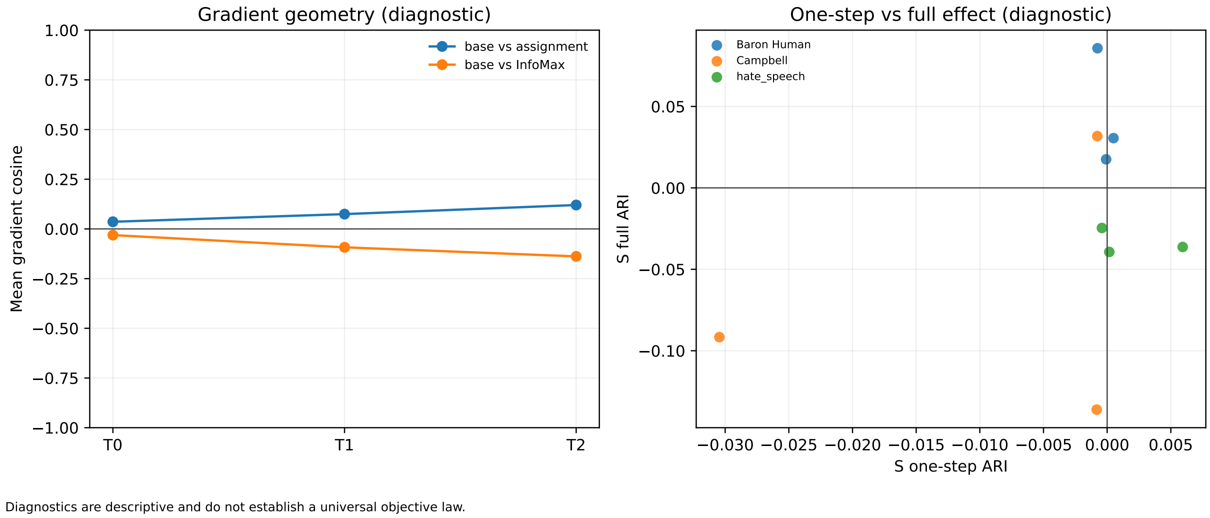


Figure 4: E2 diagnostic geometry. Gradient cosines and one-step/full effects are descriptive records; they are not a universal objective-compatibility law.

post-hoc label-aware descriptors; E3 had no eligible replay rows, and neither stage proves objective conflict, readout failure, or a universal local-to-global law. Fourth, the independent holdout is zero of six and `inconclusive_not_completed`, so independent validation remains unestablished; the frozen candidate/adapter manifest and its pool shortfall were fixed before outcomes, while the recorded CUDA failure is only incomplete compute. Finally, A2 retained E1 but prohibited E4, new Gates/losses, and V26 or rescue routes; no new architecture or comparison was used to make the narrative more attractive.

Conclusion. Structural information is not synonymous with clustering utility. The audited history provides an observational atlas of heterogeneous intervention outcomes, and a matched V21 case study shows that topology-dependent selection can be positive or negative across data sets even when generic intervention and selection noise are controlled. The defensible contribution is therefore not another Gate; it is a mechanism-localization and claim-firewall protocol that identifies where structural information may be lost and states clearly what the current evidence does not establish.

A Evidence and audit boundary

The paper-facing bundle is an analysis-only export from frozen V25 artifacts. It does not retrain a model or treat rows, seeds, or coordinates as independent population units. The exact source files and SHA256 values used to generate the tables are recorded in `tables/latex_assets_manifest.json`. The full per-job contract, branchpoint, schedule, label-isolation, and incomplete-compute records remain in `result/V25_systematic_mechanism_study/`.

The scMAE source is not cited because its local PDF has not passed the repository citation lifecycle. This is an explicit provenance boundary, not a substituted reference. The paper also does not report any independent holdout result: the frozen holdout status is `inconclusive_not_completed`.

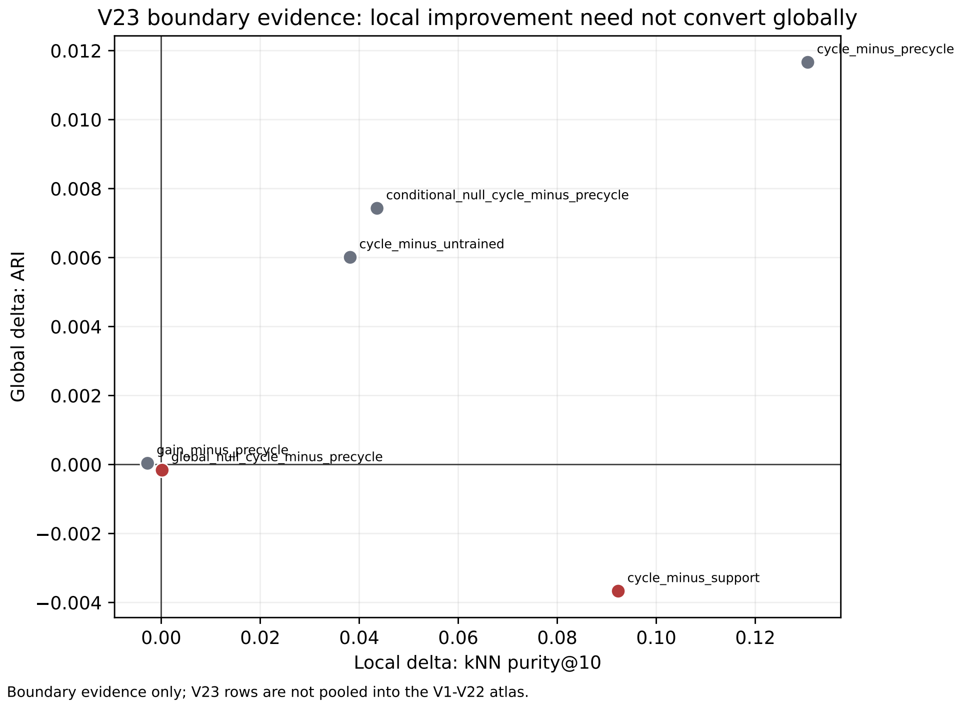


Figure 5: V23 local/global boundary evidence. These rows are kept outside the V1–V22 atlas and are not a universal theorem.

References

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